

Raising EPC Standards: A Triple Win Strategy for the UK Housing Market and Reaching Net-Zero

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Abstract

This paper employs a microeconomic approach to estimate the impact and cost of imposing Minimum Energy Efficiency Standards (MEES) of Band C on the housing sectors in England and Wales. Using an empirical investigation that leverages historical transaction prices, energy performance certificates (EPCs) from 2008-2022, and active listings from 2016-2022, the study estimates an average XXX£ per household, up to £XXX nationwide. While the study suggests property owners could capture green premiums from increased energy efficiency at point of sale, which may compensate some of the large upfront costs associated with retrofitting to a higher standard, we estimate that up to 500,000 privately rented dwellings may be at risk of not being unable to comply with the new policy. Using listing data to empirically measure the effect of such a change in supply in the market, we conclude that a sell off could potentially be leading to an 11% increase in housing affordability. This would suggest that harnessing market dynamics to achieve net-zero homes by 2050 through higher energy efficiency standards could result in increased efficiency and lower carbon footprint, increased affordability, and a reshuffling in property ownership.

Keywords: Energy efficiency, Green premium, Housing market, Net-zero, EPC standards, MEES, UK housing policy

1. Introduction

The UK's housing sector is at a critical crossroads, facing a multitude of challenges that are deeply intertwined with the nation's broader socio-economic and environmental goals. Among these challenges, the need to decarbonize the housing stock to meet Net Zero commitments by 2050 stands out as a pivotal issue. A significant portion of the UK's residential buildings, particularly those in the rental sector, are currently underperforming in terms of energy efficiency since as of 2022, homes with an Energy Performance Certificate (EPC) rating above Band C only represented 42% and 37% of the total stock in England and Wales, respectively [1]. This underscores the vast potential—and necessity—for improvements in the energy performance of the UK's housing stock.

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In the 2024 elections' manifesto, the Labour Party mentioned its intentions to invest an additional £6.6 billion over the current parliament to increase home energy efficiency as part of its 'Energy Independence Act' [2] as part of their plan to take increasing actions to reduce energy poverty and increase UK home energy efficiencies in their pursuit of achieving NZ [3]. This follows decades of multiple policy agendas that have been unsuccessful at engaging privately rented dwellings to improve their energy efficiency [4]. In October 2021, the UK Government released the Heat and Buildings Strategy, outlining its approach to decarbonizing homes and buildings [5]. A primary objective of this strategy is to enhance the energy efficiency of as many homes as possible across England and Wales, with the goal of achieving an EPC rating of Band C by 2035, provided this is practical, cost-effective, and financially viable. Additionally, the strategy aims for as many fuel-poor homes as reasonably practicable to reach an EPC Band C by 2030. While there are existing minimum energy efficiency standards (MEES) for privately and social-rented homes in the UK and Wales [6], new and existing private tenancies were required to be certified as Band E or higher since April 2018 and 2020, respectively (The Energy Efficiency (Private Rented Property) (England and Wales) Regulations, 2015). Should landlords fail to achieve MEES, they may be required to pay a fine up to £5,000 [7]. While the UK government does offer grants to partly cover the costs of retrofitting a dwelling up to standards, such as the Boiler Upgrade scheme which subsidises the deployment of heat pumps, and no property owner is currently obligated to spend in excess of £3,500 on reaching the MEES [7]. In other words, when a property is unable to reach the MEES within the given expenses, landlords must only make improvements up to that amount [7]. In September 2020 and January 2021, the government consulted whether to amend the current MEES to Band C by 2025 for new leases and 2028 for existing ones and revise upward the £3,500 threshold [5]. In September 2023, then Prime Minister Rishi Sunak announced that the target for UK rented homes would be scratched, which was later brought back by the Labour Party [2].

Retrofitting homes has been referred to as the most significant challenge to meet the UK Net Zero goals [8, 9]. As with all infrastructural transformations, sustainable structural retrofits are difficult to implement at a rapid and large scale due to structural, technical and behavioural path dependencies [10]. Committing to green retrofits to limit energy consumption and resulting carbon emissions has been not only historically difficult [11, 12], but also complicated due in part to UK property law which acts as an additional barrier for dwellings with mixed tenures and leaseholders (particularly flats) as it requires consent from all stakeholders involved [13, 14]. This is further exacerbated by properties' long lifetime and slow retrofitting rates, making it difficult and costly for the housing sector to shift from its high-emitting paradigm, as it is estimated that nearly 80% of properties were built to standard levels below those required to meet Net Zero targets [15, 9].

While incentives to commit to expensive retrofits are mixed, there is evidence that mandatory energy efficiency disclosure have resulted in green premiums, increased energy investments, and helped correct some market failures in the housing sector [16, 17]. Green Premiums in particular have been extensively studied as consumer's willingness-to-pay for homes with higher energy efficiency at the time of purchase [18, 19, 20, 21, 22, 23] or rent [24, 25, 26]. While most empirical results concur green premiums on rent and sales prices to be positive (usually around 6% and 7.6%, respectively), there are mixed interpretations about the cost-effectiveness of undergoing energy efficiency improvements for landlords [20]. This is especially the case for deep retrofitting costs which may have long and unpredictable pay-back periods, at the time

of resale. Nearly 28% of all property owners refusing to retrofit do not do so because of the high costs of retrofitting [1]. It is important to note that estimating these green premiums suffers from confounding factors such as increased demand for homes outpace the supply of newly built homes in the UK [27, 28], and this widening supply gap exacerbates housing shortages, contributing to rising property prices [29]. COVID-19 has also exacerbated the housing crisis, as demand for bigger properties which could accommodate for remote working increased, even well after the pandemic [30]. The resulting housing instability and inaccessibility have far-reaching implications, affecting socio-economic, political, and health systems [31, 32, 33, 34]. A notable consequence of this crisis is the decline in housing mobility, particularly among younger generations, which has led to a sharp increase in inequality and reduced disposable income [35]. This trend is expected to persist, further deepening disparities in the housing market and willingness to commit to green retrofits [29, 36].

In parallel with the housing supply crisis, the private renting sector has expanded rapidly over the past two decades [37, 38]. As of 2022, approximately 4.6 million households, or 19% of tenures, were in the private rental sector in England [39]. Despite the recent trends in unaffordability, there is still an ever-increasing demand for properties as people continue to want to own their own homes [27]. A significant proportion of these private renters—over 62%, equating to around 2.8 million households—expressed plans to buy a home in the future, a notable increase from 56% in 2019/20 [39]. Private renters often face additional challenges, including higher costs for utilities as over half of private renters pay their gas bills via direct debit, and two-thirds do the same for electricity (EHS, 2022). The most common type of tenancy in the domestic sector is a short-hold tenancy (EHS, 2022) leading to increased insecurity amongst the younger and lower-income groups [34]. Moreover, private renters generally live in homes with poorer energy performance compared to social renters, further compounding their financial burdens and quality of living (EHS, 2022).

Overall, the residential sector is a significant contributor to the UK's energy consumption and carbon emissions since UK households consume on average 3,500 kWh of electricity and 12,000 kWh of gas annually [40]. A substantial portion of these emissions stems from heating and electricity use within homes, which are primarily driven by fossil fuels. Notably, space heating accounts for about 63% of household energy consumption, while water heating, lighting, and appliances collectively contribute to the remaining 37% [41]. Energy prices have fluctuated considerably in recent years, with significant increases observed during periods of geopolitical instability, such as the Russian invasion of Ukraine which exacerbated energy price volatility as Europe faces simultaneous disruptions in gas supply and a shift towards alternative energy sources, including renewables (**Drax Group, 2023**). The penetration of renewables in the UK grid has been a crucial factor in buffering these price shocks, with renewables accounting for 40% of electricity generation in 2022, up from 29% in 2017 (**National Grid ESO, 2023**). Retrofitting strategies have shifted from energy conservation measures (i.e., improving wall insulation and double-glazed windows) towards integrating smart technology measures to enhance energy efficiency and integration of renewable technologies [42]. Space heating and hot water are responsible for most of the energy consumption in UK households and have hence been considered promising avenues for end-user decarbonization [11, 43]. Despite being a hard-to-abate sector, reducing end-use emissions associated with heating and powering the domestic building stock seems the most cost-effective measure [43, 44]. Technical predictive scenarios have suggested the necessary CO₂ emission reductions could be achieved through technology readily available when

accompanied by the decarbonization of the system-level electric grid and increasing switch away from fossil fuels [11, 43, 45]. However, the current retrofitting rates are far below the expected levels to reach NZ targets. Hence, there is an increasing need for policies to enable technological uptake within domestic dwellings if the UK is to meet its net zero targets [46, 45].

Heightened concerns regarding the net effect of climate change on domestic energy usage has also led to increased interest in energy conservation and preventative measures to mitigate the anticipated rise in insulation and air-conditioning needs [47, 48]. In these scenarios, financial motivations, particularly the payback period and projected energy savings, often dominate the decision to renovate, suggesting a "double-win" strategy that reduces both energy costs and carbon emissions [49]. The effectiveness of energy efficiency labels to affect property prices has been shown to vary by climatic conditions [50] and age of target audience [51], further adding to the heterogeneous and unpredictable returns from the perspective of property owners. The cognitive perceptions and financial situation of homeowners may significantly impact what is considered a cost-effective energy renovation [52, 53, 54]. Overall, for homeowners to commit to green retrofits, their preferences and values must align with their perceived financial returns on investment, in that they must be able to expect either an increase in the future sale price and/or a reduction in operational costs (i.e., energy bills) over a sufficiently short pay-back period [54, 55], which results in a demographically and geographically heterogeneous decision-making landscape [56].

This study contributes to the ongoing discourse on housing policy and energy efficiency by providing a nuanced understanding of how imposing EPC standards might influence the UK housing market. This research offers insights into how such regulations could not only help achieve environmental targets but also address housing supply and affordability issues. Through detailed empirical analysis, this study aims to inform future policy decisions, ensuring that the UK can meet its Net Zero goals while also fostering a more equitable and efficient housing market.

The contributions to the literature are threefold. First, this study provides an empirical estimation of the costs of retrofitting and their associated impact on energy consumption, carbon emissions and overall energy rating. Second, this study attempts to quantify the green premium associated with higher energy ratings as stated in their Energy Performance Certificate (EPC) at point of sale using transaction data. Third, the study estimates the potential effect of an increase on the supply side on property prices using listing data.

The remainder of this paper is structured as follows. The next section describes the different datasets used in this study. The third section presents the methodology alongside the main results. Section 4 concludes.

2. Data

Empirical data for this study was collected from three sources, including two open datasets that are historical records of transactions in the UK¹ and Energy Performance Certificates (EPCs)

¹HM Land Registry: *Price Paid Data*. (2016). [Dataset]. <https://www.gov.uk/government/collections/price-paid-data>

of private domestic dwellings in the UK and Wales²; as well as aggregated listing data provided by Zoopla.³ The first two datasets were consolidated and matched by unique property identifiers, called Unique Property Reference Numbers (UPRNs), and two mapping files were downloaded from the University of Glasgow Urban Big Data Centre (UBDC), namely the Price Paid Data to UPRNs⁴ and the EPC Individual lodgement identifier (LMK) to UPRNs⁵. The linkages above provided an accuracy rate of 96% [57]. Outliers occurred where the parent UPRNs (e.g. entire building with a unique address) were not distinct from their child UPRNs (e.g. flats in the building) which resulted in anomalistic amounts of transactions. These occurrences were removed during the data cleaning. The data processing for this study was conducted using RStudio, an integrated development environment (IDE) for R. RStudio was chosen due to its comprehensive suite of tools for data manipulation, statistical analysis, and visualization.

2.1. Energy Performance Certificates

Energy Performance Certificates (EPCs) were introduced in the UK on 1 August, 2007 by the Housing Act 2004 and EPBR 2007.⁶ Figure 1 below represents the energy efficiency from a typical certificate processed for this study. EPCs are determined for a dwelling upon

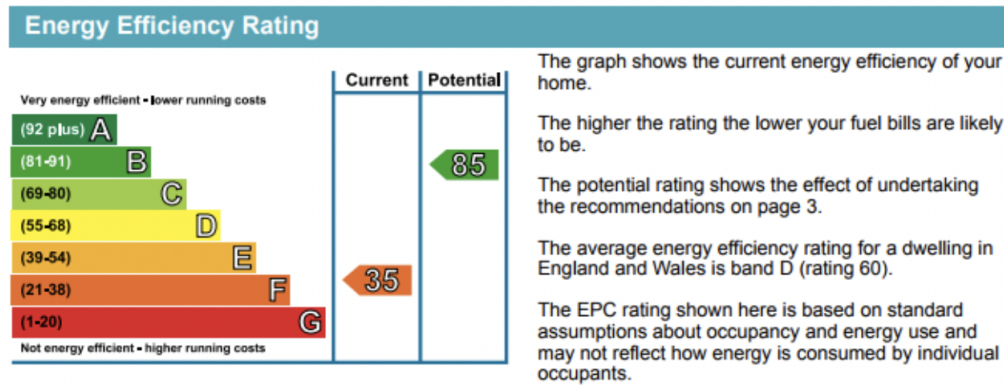


Figure 1: Typical example of Energy Efficiency Rating reported in an EPC.

inspection using the Standard Assessment Procedure (SAP). The SAP and RdSAP (Reduced SAP) methodologies are developed by an external organization - the Building Research Establishment (BRE) - and have been criticised for estimating the cost-efficiency - rather than energetic performance - of buildings [58]. Since SAP ratings mostly take into account energy appliances and house geometry (as estimators of energy use) [59, 60] they can overlook the 'fabric' energy efficiency

²Energy Performance of Buildings Data England and Wales. (n.d.). [Dataset]. Retrieved 16 August 2024, from <https://epc.opendatacommunities.org/>

³Zoopla Limited, © 2024 2. Zoopla Limited. Economic and Social Research Council. Zoopla Property Data, 2024 [data collection].

⁴Urban Big Data Centre (2023). Price paid data to UPRN lookup [Data set]. University of Glasgow. <https://doi.org/10.20394/agu7hprj>

⁵Urban Big Data Centre (2023). Energy Performance Certificate to UPRN lookup [Data set]. University of Glasgow. <https://doi.org/10.20394/okfrrz60>

⁶The Energy Performance of Buildings (Certificates and Inspections) (England and Wales) Regulations 2007

[61]⁷ of buildings and may hence not act as the most efficient predictors of carbon emissions [62]. Studies comparing smart meters and domestic EPC energy use ratings have also shown discrepancies in estimations, specially significant over-predictions of energy use in lower rated dwellings due to assumptions on building characteristics and occupancy in the SAP modelling [63]. Other findings suggest this 'measurement error' results in less-accurate estimations in Bands C and lower due to the increase variance in repeated measurements [64]. Energy-poor properties tend to be the most significantly impacted by the lack of consistency and accuracy of EPC ratings. To avoid these errors, some studies restricted their sample to Bands D and above [21] when attempting to estimate consumer's willingness to pay for energy efficiency. However, this study aims to explore the inequalities and aggregate impacts of improving the housing sector to their maximum potential energy efficiency, which inevitably will affect the lower-rated dwellings. Additionally, when estimating the green premium (see 3.3) of green-labelled dwellings, it has been shown that government endorsement, simplicity, and financial incentives are key factors determining the trust in certificates rather than the accuracy of the latter [65, 66, 67]. Therefore, while there are shortcomings to the SAP methodology, its wide application serves a consistent and standardized index of energy use, sufficient for the purpose of investigating market responses to consumer awareness and policy-incentives.

The energy efficiency ratings of 15 million dwellings were extracted from Energy Performance Certificates (EPC) available on the Energy Performance of Buildings Register ('the register'). Since 2008, all properties listed for sell, rent, or newly constructed must undergo an inspection [59]. A certified assessor assigns an energy efficiency ratings from the Standard Assessment Procedure (SAP) in newly built homes and Reduced SAP (RdSAP) in retrofitted homes [68, 69]. SAP energy ratings are determined from an estimated ratio of energy consumption and production for each property and provide an energy cost factor on a scale of 1 to 100 (or above 100 if a site generates more energy than it needs) [60, 61]. The given numerical scores can be converted to ABCDG bands of energy efficiency - from least efficient and high running costs (1-20 : G) to most energy efficient and low running costs (92+ : A). EPCs provide additional information on the insulation and efficiency of all elements of the property, heating methods, estimated energy needs/costs, and carbon emissions, as well as more detailed information about the properties (such as property size, building age and other property characteristics). After selecting for the observations that could improve their EPC and cleaning the dataset, 15,602,800 dwellings were kept and analysed as they were able to reach the minimum EPC rating standards imposed by the PRS.

2.2. Price Paid Data and Listing Dataset

The HM Land Registry's Price Paid data provides nearly a full record of all residential property sales in England and Wales since 1995.⁸ Transactions excluded from the Price Paid Data are homes not registered with the HM Land Registry and all unorthodox transfer of property rights (i.e., Deeds, 'Right to Buy' discounted properties, etc.). While previously only commercially available, the UK Land Registry data became accessible to the public as of 2013, making historical records of cash and mortgage transactions open data. For the purpose for this study, variables included the transaction prices, date, location (by postcode), property type, and UPRNs. In total

⁷refers to passive characteristics of buildings which affect energy efficiency, such as insulation

⁸HM Land Registry: Price Paid Data. (2016). [Dataset]. <https://www.gov.uk/government/collections/price-paid-data>

14.5 million transactions were extracted from the Price Paid data between 2007-23, of which 9.6 million consisted of unique addresses (or UPRNs).

Table 1: Summary Statistics of average transaction price by property type and EPC rating as defined by RdSAP.

EPC Rating	Bungalow	Flat	House	Maisonette	Park home
A	297144.7	328078.0	366825.9	384231.6	-
B	298159.8	297840.0	304817.2	342341.8	-
C	279565.3	228881.0	264755.1	254608.2	278219.6
D	264678.9	243516.0	264634.6	273549.8	184001.2
E	274003.5	238571.2	283038.0	269317.8	246575.5
F	277801.3	211757.1	299006.1	235927.6	226707.8
G	256178.3	206546.6	240768.1	218924.4	291646.8

Transaction prices were adjusted for inflation using the Retail Price Index (RPI) for the sample period which was sourced from the Office of National Statistics (ONS) ⁹. While the Consumer Price Index (CPI) has been the prevailing inflation index in the UK since 2003, after its inception in 1996 [70], both indicators rely on the overall change in price paid by households for the goods and services they consume [71]. However, they differ in the 'basket of goods (data) and formula used [72]. As seen in Figure B.7 in Appendix, the RPI relies more heavily on housing expenditures (e.g., housing purchase costs, mortgage interest payments, building insurance, etc.) making it more sensitive to housing market conditions [72]. This is especially relevant during the subprime mortgage crisis, as the RPI provided a significantly better representation of the turbulent housing prices therefore, the "real" transaction prices were adjusted from the RPI as of January 2019. Properties in the top 1% of transaction prices were considered luxury goods and hence removed from the panel of observations since they are subject to distinct consumer behaviours [73] which may directly affect the willingness to pay for certain characteristics and services such as green labels [74, 75, 76] hence affecting the underlying trends investigated here.

Finally, access to rental and sales Zoopla property data from 2016-22 were granted by Zoopla Limited¹⁰ under licensed agreement and aggregated at postcode level each property in order to control for variations in supply in the housing market across the UK and Wales. For each property, the time period listed on the market until sold or rented was provided enabling the computation of the active listings in an area surrounding a point of transaction. For each listing period, the number of listings was therefore computing across district areas and added as controls for supply. Figure 3 provides an overview of the supply of private homes for rent and sale as aggregated therein, as measured in the number of active listings in a given area and month, showing clear evidence of seasonal trends.

⁹<https://www.ons.gov.uk/economy/inflationandpriceindices>

¹⁰Zoopla Limited, © 2024 2. Zoopla Limited. Economic and Social Research Council. Zoopla Property Data, 2024

3. Methods and Results

In this section we explore ...

3.1. Costing Energy Retrofits and their impacts

Dwellings with at least one inspection recording an EPC (15.6 million registered), were analysed. Each EPC records current and potential energy efficiency, rating, retrofits available together with recommendation file which lists potential retrofits in a given property.

Should the property be able to raise their EPC, the record will include the potential energy rating alongside their current energy rating, respectively denoted R_p and R_c in the equation below, so that the difference in energy efficiency is explained by committing to specific retrofits (with R_i a list of retrofits):

$$E_{i,p} - E_{i,c} = f(R_i)$$

We start with estimating the impact of distinct energy retrofits on a series of dependent variables, including Energy Efficiency, energy consumption, costs and emissions in a given dwelling. We run the following regression using Ordinary Least Square (OLS):

$$Y_i = E_{i,p} - E_{i,c} = \beta_0 + \sum_{m=1}^{63} \beta_m X_{i,m} + \gamma_i \cdot time_{fe} + \epsilon_i \quad (1)$$

where Y_i captures Energy Efficiency (score out of 100), Yearly energy Consumption (kWh/m^2), costs (£ kWh) and CO_2 emissions. $X_{i,m}$ is a vector of energy retrofits in a given property i . The environmental impact ratings are based on the carbon emissions of buildings per square meter [6]. Time fixed effects are captured by $time_{fe}$ and ϵ_i is the error term.

Results from the five OLS regressions (Eq. 1) are presented in Table 2 below. Coefficients are all significant (p-value less than 0.0001). They provide insight into the marginal effects of each completed structural retrofit on the energy efficiency score, energy consumption (in kWh and £), and emissions (in CO_2 and impact score) of dwellings. According to table 2, installing a wood pellet stove with boiler and radiator (ID 23 & 39) can increase a property's energy efficiency rating by 22.79, making this improvement option the most optimal decision when aiming to increase the EPC score (Y_1). As expected, improvement ID 23&39, also yields the greatest impact with all other dependent variables, resulting in the most impact retrofit available to homeowners from a purely results-based perspective. When comparing across all the regression results, it seems the most 'impactful' renovations involve improving a dwelling's heating systems (e.g., Fan assisted storage heaters, gas condensing boilers, replacing heating units, etc.). This is consistent with prior findings that the majority of residential energy consumption and associated emissions stem from space and water heating [11, 43]. Conversely, the recommendations associated with the lowest impact estimates from the models, are mostly associated with fabric energy efficiency and energy conservation measures (i.e., loft insulation, double window glazing, party wall insulation, etc.). The results were appended and used during the subsequent parts of the analysis as indicators of impacts for each dwelling.

3.2. Cumulative costs and environmental impacts of reaching EPC-C

As previously mentioned, property owners are not obligated but highly encouraged to follow the ordered improvement suggestions on the EPC [59]. Figure 4 provides an overview of the 3

Table 2: Regressions Results: Impacts of Improvements according to EPCs 2008-22

Independent Variables		Dependent Variables					Cost (£)
		EPC(%) (Y ₁)	Consumption (kWh) (Y ₂)	CO ₂ (Y ₃)	Environmental (Y ₄)	Bill (£) (Y ₅)	
1-3	Insulate hot water cylinder with 80 mm jacket	3.66***	-40.56***	-0.37***	3.93***	-62.25***	22.5
4	Hot water cylinder thermostat	4.48***	-32.83***	-0.66***	4.78***	-121.14***	300
5	Increase loft insulation to 270 mm	3.20***	-26.05***	-0.45***	3.74***	-93.96***	225
6	Cavity wall insulation	4.79***	-40.68***	-0.64***	5.48***	-126.39***	1000
7	50 mm internal or external wall insulation	6.30***	-52.08***	-0.89***	7.01***	-186.02***	9000
8	Replace single glazed windows with low-E double glazing	2.31***	-21.55***	-0.45***	2.22***	-90.07***	4900
9	Secondary glazing to single glazed windows	2.44***	-12.72***	-0.66***	1.26***	-110.11***	1250
10	Draughtproof single-glazed windows	4.46***	-28.88***	-1.01***	3.99***	-198.74***	100
11-18	Upgrade heating controls	2.29***	-15.11***	-0.32***	2.60***	-62.79***	400
19	Solar water heating	2.84***	-30.54***	0.47***	3.20***	84.64***	5000
20-21	Replace boiler with new condensing boiler	4.39***	-33.06***	-0.75***	5.53***	-139.94***	2550
22	Replace boiler with biomass boiler	14.62***	-106.26***	-11.78***	57.12***	-585.71***	10000
23&39	Wood pellet stove with boiler and radiators	22.79***	-234.00***	-14.62***	58.98***	-1161.85***	10000
24&30	Fan assisted storage heaters and Dual Immersion Cylinder	9.19***	-58.68***	-0.75***	4.89***	-179.76***	1123.8
25&31	Fan-assisted storage heaters	9.19***	-58.68***	-0.75***	4.89***	-179.76***	998.75
26	Replacement warm air unit	2.38***	-14.73***	-0.38***	3.00***	-95.81***	1875
27,29,32,40-42	Change heating to gas condensing boiler	18.36***	-173.60***	-2.20***	17.87***	-586.96***	5000
28&43	Condensing oil boiler with radiators	10.24***	-18.21***	-2.72***	7.53***	-429.72***	12500
34	Solar photovoltaic panels, 2.5 kWp	6.41***	-27.92***	-1.47***	5.01***	-145.52***	4500
35	Low energy lighting for all fixed outlets	0.88***	-5.38***	-0.11***	0.82***	-35.12***	30.40
36	Replace heating unit with condensing unit	2.92***	-22.77***	-0.40***	3.50***	-83.92***	2600
37-38	Install condensing boiler	13.65***	-47.16***	-3.99***	11.48***	-680.93***	2600
44	Wind turbine	6.20***	-37.59***	-0.87***	4.66***	-70.00***	19999.3
45	Flat roof insulation	4.05***	-35.29***	-0.60***	4.49***	-132.82***	1175
46	Room-in-roof insulation	7.57***	-49.07***	-1.92***	8.32***	-398.64***	2100
47	Floor insulation	5.41***	-33.05***	-0.63***	6.38***	-88.26***	1000
48	High performance external doors	5.07***	-50.21***	-0.22***	3.50***	-77.32***	1365.6
49	Heat recovery system for mixer showers	1.49***	-50.21***	-0.09***	1.01***	-50.43***	655
50	Flue gas heat recovery device in conjunction with boiler	3.81***	-42.05***	0.56***	2.85***	80.30***	650
56	Replacement glazing units	1.49***	-16.67***	-0.15***	2.24***	-29.63***	1200
57	Suspended floor insulation	4.01***	-32.61***	-0.58***	4.65***	-114.17***	1000
58	Solid floor insulation	4.44***	-34.68***	-0.64***	4.62***	-112.41***	5000
59&61	High heat retention storage heaters and dual immersion cylinder	17.32***	-85.81***	-0.72***	4.21***	-489.40***	1501.3
60&62	High heat retention storage heaters	9.14***	-64.51***	-0.49***	3.55***	-249.02***	1346.1
63	Party wall insulation	2.53***	-32.71***	-0.14***	3.30***	-24.36***	450
Year Fixed.E		yes	yes	yes	yes	yes	
Efficiency Gradings		yes	yes	yes	yes	yes	

NOTES: These estimates are the results of OLS regressions. The table reports the impacts of each improvement (independent variables). The coefficients represent the change in the dependent variables per improvement completed (independent variable). The columns report the estimated impact of each improvement on the change in EPC(Y₁), change in Energy Consumption(Y₂), CO₂ emissions (Y₃), Environmental Impact factor (Y₄), and total energy costs in (Y₅). Regressions include household fixed effects, along with time controls in the second column. Robust standard errors in parentheses. Significance levels are indicated as *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

'retrofitting strategies' used to model the complex decision-making process of property owners

willing to retrofit. Faced with the EPC recommendations lists, owners can opt for the 'low hanging fruit' improvements if faced with requirements to retrofit and low financial capabilities [77]. In this strategy, it is assumed the cheapest structural retrofitting suggestions would be firstly implemented. Conversely, the strategy 4 would highlight dwellings willing to invest first in improvements that could yield the highest impact (change in EPC score) to retrofit with minimal interventions. Finally, a more complex scenario is explored (strategy 3). In this model, it is assumed homeowners could estimate the 'highest impact per pound' improvements and preferably selected such impactful and worth-while retrofits.

The cumulative costs of achieving the EPC cap was determined for each nuanced retrofitting strategy: 1) per EPC recommendation, 2) as per EPC impact potential, 3) cheapest improvement option) and 4) by best impact per investment (EPC_impact/Cost of retrofit). The program ran on the 9.2 million dwellings that were able to rise their EPCs and had an original energy efficiency below that of the MEES amendment suggested of 69% (C-Band). The linear estimation models overestimated the cumulative and individual costs and impacts of reaching the minimum EPC rating standards (Table 3). This was probably induced as the linear estimation included some costly improvements at the of the EPC recommendations which would have yielded more than the necessary scores. Conversely, the computationally-determined costs using different 'strategies' stopped keeping track of structural improvements once households reached the set threshold. The total costs for retrofitting all able registered properties in England and Wales to an EPC rating of C range from £59 billion to £94 billion, allowing the sector to reduce its yearly emissions by 18.1 - 20 million tonnes of CO_2eq . In 2023, residential emissions from buildings in the UK accounted for 51.7 million tones of CO_2eq ¹¹. The cumulative costs would be estimated at about £82.5 billion if all home owners followed and implemented the list of recommendations provided by their EPCs. For each investment strategy, the cumulative change in energy consumption for the housing sector is also displayed and presented in GWh/m^2 . While the EPC retrofitting strategy does not yield the greatest decrease in emissions, SAP recommendations allow for the most significant decrease in energy consumption.

Table 3: Cumulative Costs for EPC Cap 69

	N. Retrofits	Household (£)	CO2 ($MtCO_2eq/yr$)	Env($MtCO_2eq/yr$)	Energy (GWh/m^2)	Total Cost (£)
By Yield	3	6597	-18.6	124.4	-881.6	60,887,982,229
By Price	3	6670	-18.3	124.6	-885.2	61,558,192,060
By EPC	3	9058	-16.1	125.3	-918.3	83,179,781,006
Linear Est.	-	9705	-12.8	103.6	-739.1	89,649,163,609
By Impact	2	9820	-20.6	115.5	-806.5	90,633,857,958

NOTES: Results were determined by accumulating retrofitting costs for all houses in need and capable of reaching an energy performance score of 69. The rule-based program ran on 9.2 million dwellings, following the EPC Recommendations (Strategy #1 in Figure 4).

According to the table therein, the "cheapest" aggregate option to reach an EPC of 69 is achieved when property owners opt in for the improvement that can yield the highest impact on energy efficiency for the lowest price. This option would require significant investments in Solar

¹¹2023 Provisional figures from ONS and DESNZ

Photovoltaic Panels (nearly 4 million across the UK) and over £13 billion national investments for external wall insulations (see table ??). Following this decision mode, reaching an EPC of 69 would require households to spend £6597 on retrofitting costs on average which would present significant savings when compared to the order recommended by the EPC assessments. Indeed, to achieve an EPC of 69 following the SAP recommendations, the average household would have to spend nearly £9000 on retrofitting costs, most of which would require investment in PV panels and solar water heating. See Appendix Appendix B for other improvement strategies. This confirms [78] study which suggested applying more complex and sophisticated methods of selecting structural improvements result in cheaper and more efficient outcomes. While the low-hanging fruit strategies (by yield or by price) provide the most cost-effective means of achieving MEES, they do not effectively reduce CO_2 emissions as much as strategies focused on impact. Similarly, following the EPC recommendations result in less emission reductions but yield the highest reduction in energy consumption (expressed in GWh per area in Table 3). These observations confirm [58] suggestions that the SAP methodologies and recommendations may create perverse incentives to reduce energy consumption rather than limit environmental damages.

While the average reported costs for the each retrofitting strategies are specified in table 3, the uni-dimensional values do not capture the heterogeneous costs across difference properties. The density plot (Figure 5) reveals the variation of house price around the mean reported cost following the EPC recommendations (red-dashed line). The multi-modal distribution of retrofitting costs seem to suggest costs are grouped into multiple price ranges - potentially indicating groups of similar property characteristics. Retrofitting costs seem to be similar across price ranges. The majority of homes likely require low retrofitting costs while the long-tail suggests certain outliers require expensive investments to achieve the minimum energy efficiency standards - most of which seem to be higher priced homes (500k+). These tails persisted despite excluding the top 1% most expensive properties to retrofitting. Out of nearly 4.45 million dwellings analysed, 60.95% of those able to reach a minimum EPC score of 69 endure retrofitting costs less than £10,000 to retrofit, while only 28.39% fall below £3,500 - according to the EPC retrofitting strategy.

While an estimate 15.6 million registered dwellings were considered 'able to improve their energy efficiency' according to their EPC ¹², 6 million observations had above-standard energy efficiencies and 1 million properties were unable to every reach a minimum C-Band energy rating (EPC Score of 69) as their 'potential energy efficiency' fell below such threshold. According to the 2015 Energy Efficiency Regulations, dwellings unable to meet MEES require to retrofit until reaching full energy efficiency potential or spending above the a maximum retrofitting budget set by the government. Retrofitting costs the all properties were then computed analogous methodologies and summarized in Table 5.

3.3. *Green premium for energy efficiency*

Green premium: 3 confounders (demand, supply and renovations/condition of the house at time of sale) we add interest rates, and listing data to take care of most important

¹²A dwelling considered 'able to improve' here refers to EPCs including improvement recommendations and a differing potential energy efficiency as outputted by the SAP software.

Table 4: Total Needed Improvements to Reach EPC C ordered by Total Costs, according to 'Highest Yield' decision strategy.

ID	Count	Description	COST (£)	ΔEPC	Energy(KWh)	CO2 (CO ₂ eq/yr)	TOTAL COSTS (£)
34	3922456	Solar photovoltaic panels, 2.5 kWp	4500	6	-1.469	-27.922	17651052000
7	1449750	50 mm internal or external wall insulation	9000	6	-0.886	-52.080	13047750000
20	2247525	Replace boiler with new condensing boiler	2550	4	-0.752	-33.063	5731301126
58	833332	Solid floor insulation	5000	4	-0.636	-34.684	4166660000
19	801603	Solar water heating	5000	3	0.474	-30.535	4008015000
44	156729	Wind turbine	19999	6	-0.870	-37.588	3134470290
6	2133786	Cavity wall insulation	1000	5	-0.641	-40.683	2133786000
57	1851908	Suspended floor insulation	1000	4	-0.583	-32.612	1851908000
47	1565764	Floor insulation	1000	5	-0.630	-33.051	1565764000
8	262302	Replace single glazed windows with low-E double glazing	4900	2	-0.451	-21.554	1285279800
11	2874681	Upgrade heating controls	400	2	-0.319	-15.115	1149872400
46	498424	Room-in-roof insulation	2100	8	-1.919	-49.068	1046690400
27	201053	Change heating to gas condensing boiler	5000	18	-2.203	-173.602	1005265000
45	463028	Flat roof insulation	1175	4	-0.604	-35.291	544057900
5	1850553	Increase loft insulation to 270 mm	225	3	-0.452	-26.047	416374425
59	255465	High heat retention storage heaters and dual immersion cylinder	1501	17	-0.724	-85.813	383529604
60	254942	High heat retention storage heaters	1346	9	-0.489	-64.509	343177426
48	169938	High performance external doors	1366	5	-0.215	-50.209	232067333
50	276928	Flue gas heat recovery device in conjunction with boiler	650	4	0.562	-42.050	180003200
35	5658145	Low energy lighting for all fixed outlets	30	1	-0.108	-5.385	172007608
4	559005	Hot water cylinder thermostat	300	4	-0.663	-32.833	167701500
56	88441	Replacement glazing units	1200	1	-0.146	-16.667	106129200
30	85158	Fan assisted storage heaters and dual immersion cylinder	1124	16	-0.754	-67.301	95700560
23	7706	Wood pellet stove with boiler and radiators	10000	23	-14.619	-235.002	77060000
25	75892	Fan assisted storage heaters	999	9	-0.751	-58.682	75797135
10	620982	Draughtproof single-glazed windows	100	4	-1.012	-28.884	62098200
37	19747	Install condensing boiler	2600	14	-3.988	-47.163	51342200
63	109772	Party wall insulation	450	3	-0.142	-32.712	49397400
49	67828	Heat recovery system for mixer showers	655	1	-0.089	-7.041	44427340
26	16886	Replacement warm air unit	1875	2	-0.380	-14.726	31661250
22	2870	Replace boiler with biomass boiler	10000	15	-11.784	-106.260	28700000
1	1148041	Insulate hot water cylinder with 80 mm jacket	22	4	-0.366	-40.557	25830922
9	10160	Secondary glazing to single glazed windows	1250	2	-0.661	-12.716	12700000
43	744	Condensing oil boiler with radiators	12500	10	-2.715	-18.210	9300000
36	425	Replace heating unit with condensing unit	2600	3	-0.396	-22.771	1105000

Table 5: Dwellings of various capacities to reach the potential MEES of EPC - C.

	Already Achieved	Unable to Reach	Able to Reach EPC 69
Number of Houses	6,368,752	1,060,889	8,171,825
Total Retrofit Costs (10⁹£)	-	9.4	8.3
Reduced Emissions (MtCO₂eq/yr)	-	-1.7 to -2.0	-12.8 to -20.6
Average Retrofit Cost (£)*	-	8,571	9,058
Avg. <i>pre</i> -retrofit EPC	75	48	57
Avg. <i>post</i> -retrofit EPC	-	61	69
Average Property Price (£)	290,397	327,014	287,725

NOTE: For properties unable to reach EPC - C, retrofit costs were determined from maximum energy efficiency potential.

factors. need one more column with both interest rate and listing data. listing were aggregated/counted active per area around a property 30 days (TBD) before transaction finalised as it takes an average of 3 months to complete.

A hedonic regression model was used to reveal the marginal effects of energy efficiency and other building characteristics at the time of sale on the transaction price. Hedonic price model was used to estimate willingness to pay using transaction price data [18] using an adjusted OLS regressions of deflated prices (P_{it}) as a function of the SAP score and attributes extracted from the EPC data (X_{it}):

$$P_{it} = \alpha_i + \beta_1 \cdot \text{EPC}_{it} + \sum_{j=2}^8 \beta_j \cdot X_{j,it} + \gamma_t \cdot \text{time}_t + \epsilon_{it} \quad (2)$$

Where $\beta_{1:9}$ are the hedonic weights and X_{it} are comprised of the total floor area (X_2), property type (X_3), number of habitable rooms (X_4), extension count (X_5), building type (X_6), construction age band (X_7), and main fuel source (X_8) for each dwelling. These are included as control variables to limit omitted variables bias and improve the accuracy and validity of the model [79]. ϵ_{it} refers to the standard error terms and, due to the large amount of transactions, is expected to follow a normal distribution.

Given that the total floor area is used in the SAP methodology to determine energy efficiency ratings [60], it is suspected to be correlated with energy efficiency, there is a concern of multicollinearity and endogeneity. Such concerns are also raised for construction age as part of the key determinants of energy demand, hence ultimately reflected in SAP ratings [80]. This correlation might inflate the estimates and obscure the true relationship between each variable and transaction price. However we assume the SAP methodology to take into account this step and provide the EPC ratings as 'fitted' values from building properties. Similarly, ages of dwellings have proven to be significantly correlated to energy efficiency as most houses built after 2012 are much more likely to be energy efficient [1]. These potential sources of bias must be considered and taken into account when interpreting the regression results therein.

The hedonic price models were executed as OLS regressions using the `feols` package in R [81, 82] on 10.4 million transactions. The regression results are displayed in table 6. In line with expectations, the analysis reveals that energy efficiency is a statistically significant predictor of transaction price (p-value <0.0001). In the simple hedonic model (Table 6), the coefficient associated with energy efficiency indicates a positive impact on transaction price, with an estimated hedonic premium of approximately £486.72 for every additional improvement in the Energy improvement score. Since the average improvement in energy efficiency across all properties is of 15, the estimates therein would result in an average net green premium of £7,331.85. This suggests that homes with higher energy efficiency ratings tend to command higher prices in the housing market, likely reflecting buyers' willingness to pay [21].

When accounting for time-varying controls through the use of other fixed effects, namely monthly interest rates and number of properties listed for sale in the same region, the estimate is reduced to approximately £207.80 (Table 6). The reduction in the coefficient suggests that the initial model may have overestimated the impact of energy efficiency by not accounting for unobserved, unit-specific characteristics that do not vary over time. When controlling for the supply of housing (measured by the number of listings), the adjusted R^2 value increased significantly compared to the other models. This improvement in the model's explanatory power underscores the importance of controlling for a broad range of factors that influence housing

Table 6: Regression Results: Hedonic Price Models with Fixed Effects

<i>Independent Variable</i>	<i>Simple Hedonic Model</i> (1)	<i>Hedonic Model FE</i> (2)	<i>With Interest Rates</i> (3)	<i>With Listings</i> (4)
Energy Efficiency	500.88*** (5.54)	486.85*** (5.57)	483.26*** (5.56)	408.46*** (7.92)
Adj. R-Square	0.37	0.37	0.37	0.45
N	8,923,015	8,923,015	8,923,015	4,001,425
Total Floor Area	Yes	Yes	Yes	Yes
Property Type	Yes	Yes	Yes	Yes
Number of Habitable Rooms	Yes	Yes	Yes	Yes
Extension Count	Yes	Yes	Yes	Yes
Built Form	Yes	Yes	Yes	Yes
Construction Age Band	Yes	Yes	Yes	Yes
YEAR Time-Fixed Effects	No	Yes	Yes	Yes
MONTH Time-Fixed Effects	No	Yes	Yes	Yes
Monthly Interest Rates	No	No	Yes	Yes
Active Listings	No	No	No	Yes

NOTES: These estimates are the results of OLS regressions. The table reports the effect of energy efficiency score (out of 100) on property transaction prices (dependent variable). The coefficients represent the change (in £) in the dependent variable (price) per unit change in the independent variable (energy efficiency). The first column reports the results for the simple hedonic model without time fixed effect and the second column includes the time fixed effects. Regressions include household fixed effects, along with time controls in the second column. Columns (3) and (4) use monthly interest rates and average yearly sale listings by region as fixed effects, respectively. Robust standard errors in parentheses. Significance levels are indicated as *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

prices, beyond just energy efficiency.

3.4. Elasticity of supply

For this the sale price elasticity were parametrically estimated from the given active listings dataset¹³ from the equation below, similar to work done by [83]. Price elasticity refers to the effects of additional demand shocks on residential market prices and draws from the economic theory of "law of demand" [84].

$$\ln(P_i) = \beta \cdot \ln(\text{active listings}_i) + \gamma_i \cdot t + \text{rent}_i : t_i$$

According to the properties at risk and potential decision framework to sell or rent therein, over 500 thousand properties risk entering the housing market for sale.¹⁴ These refer to homes current renting that may be forced to sale faced with imposed MEES. This may have significant impacts on the UK housing market. According to the results (table 7) elasticity is significant and is estimated that for every additional 1% in active listings, property prices decrease by 2%. Since about 8.1 million properties were currently listed in the market (from dataset), an additional 6.3% properties might enter the market as a result of the MEES. Inducing a 11.3% drop in property prices.

¹³Zoopla Limited, © 2024 2. Zoopla Limited. Economic and Social Research Council. Zoopla Property Data, 2024 [data collection]. University of Glasgow - Urban Big Data Centre

¹⁴all properties with construction age before 1949 and in districts with high retrofitting cost and high fuel-poverty (nearly 2.6 million) and adjusted for the estimated 19% of dwellings for rent

Table 7: Regression results showing the effects of base rate and other predictors on ln of growth_price.

<i>Independent Variables</i>	House Price (£)
ln Active Sales	−0.02** (0.00)
Interest Rates	0.02* (0.01)
Ajd. Interest Rates	−0.01* (0.00)
N.	3,359,780
R ²	0.03
Adj R ²	0.01
Property Type	yes
District	yes
Year	yes
Month	yes

NOTES: These estimates are the results of OLS regressions. The table reports the effect of supply (number of active sales) on property transaction prices (dependent variable). Robust standard errors in parentheses. Significance levels are indicated as *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

While the estimates above represent an immediate effect of the induced supply shock, which will inevitably decrease with time [84], it is highly significant. The increased supply of homes may have direct effects on the local housing prices but also have knock-on effects on neighbouring regions which buyers consider substitutable [85]. This may have implications on the current housing crisis by increasing the supply of homes and potentially increase affordability across the UK.

4. Discussion

Note: similarly no impact on rental market (as rents are always at max capacity). The ogermment should focus on insulation. we also need to distinguish between all homes and rental properties when mentioning the figures below. Are the cost tables 3, 4 and 5 for all properties or just rentals? (we will need both).

Imposing a stricter energy efficiency standard (MEES) of 69 on all dwellings capable of reaching such standards has the potential of decreasing the UK's housing annual emissions by 38.2-42.5% from 2022 levels ¹⁵ (and 52.3 - 55.6% from 2017 baseline levels) [86, 87, 88]. If 8.2 million properties retrofit to an EPC of C within 2028 - as recommended by the CCC [89] - the UK housing sector would surpass its NZ targets of decreasing 50% of residential emissions by 2032 compared to 2017 [86]. The UK could be well on the road to Net Zero. While setting MEES could serve as a critical pathway for England and Wales to meet their Net Zero goals by 2050, the success of the policy amendment must be weighed against the financial incentives for households and the broader impacts on the already challenging domestic housing sector.

Households across England and Wales would have to bear a cost of £91.6 billion ¹⁶ in energy improvements to reach a minimum EPC standard of C (score of 69 out of 100) - or at least

¹⁵values determined from EPC retrofitting strategy for both houses able and unable to reach 69.

¹⁶This figure includes £82.5 to reach an EPC of C and £9.1 for all other dwellings to reach their maximum energy efficiency potential.

maximum efficiency potential with the current ceiling. Results indicate over £29 billion would be spent on wall insulation (ID 7), followed by £13 billion in solar water heating technologies (ID 19). The overall costs of these two technologies have not significantly come down in recent years, sitting at £10,000-9,000 and £5,000 - respectively. These were possibly induced by a lack of demand and 'experience learning-curve' effect [90, 91, 92] as uptake of solar water heaters and wall insulation measures had remained low in the past - despite government incentives and EPC recommendations [93]. The average retrofitting cost at household level of reaching MEES ranges from £6,434 to £10,000 based on three different retrofitting strategies. However, as with aggregate costs, such prices may be overestimated as experiential learning curves and economies of scale have not been taken into consideration [90]. Should households commit to retrofits as suggested in their EPC reports, the resulted cost would be higher than identifying the most impactful retrofits with respect to price per price.

This is concerning as property owners have limited information when deciding to commit to green retrofits and other geographical and time dependent factors are likely to influence this decision as well, including the potential rent premium for higher efficient properties or the surcharge paid for a property due to lack of supply, thus exacerbating inequalities. Landlords might simply decide to sell their rental properties as future rents would not be able to compensate for the upfront costs associated with meeting the new EPC and the loss of rents while retrofits are being conducted.

The average retrofitting costs of achieving minimum energy efficiency of 69% - or highest energy efficiency potential for less-able properties - vary widely by construction age and price range. Older properties, especially pre-war homes, would have to incur higher retrofitting costs due to their low efficiency. These account for over 35% of properties in the UK and Wales, with wales having more very old homes [94]. Property owners would have to spend £9-15k in renovations for those able to reach 69% and £8-10k for others to reach their maximum energy efficiency.

Although not highly geographically heterogeneous (Figure 6), certain districts seem to show above-average retrofitting costs to reach the new requirements. These correspond to the local authority districts of the Scottish Borders, Northumberland in the Northeast as well as Wiltshire and West Devon in SouthWest which are largely rural [95]. These align with existing reports of unusually high poverty in these districts [96, 97]. Fuel poverty is often associated with low income and less-energy efficient households as affordability for energy costs are low [98].

Although the figures therein represent the estimated costs as houses implement recommendations from the EPCs, there is no legal requirement for property owners to follow such order. Instead, a minimum energy efficiency standard - as suggested by the UK government in 2021 - would take a 'whole house retrofit' approach and not take into account the nuances of property-level decisions. However, as seen by the nuances and sensitivity of the costs by retrofitting strategy (3), an surgical approach to retrofitting would capture much more nuances to overall retrofitting costs and total abated emissions [8, 9]. Indeed, the results confirmed previous studies' criticism on the energy-cost focused narrative of the SAP and RdSAP recommendations [58]. A more sequential and tailored approach to implementing energy efficiency improvements, prioritizing those with the highest yield, could significantly reduce household costs and facilitate more coordinated investments in specific improvement industries, potentially driving down retrofitting expenses [8]. As previously discussed, the EPC retrofitting scheme designed to meet the Minimum Energy Efficiency Standards (MEES) has emphasized investments in technologies that are individually

expensive and have not demonstrated significant price reductions over time. In contrast, if households were to prioritize improvements that offer the greatest efficiency gains relative to their cost, the majority of aggregate spending would likely be directed towards the installation of solar photovoltaic (PV) panels (ID 34). Such large-scale investment—potentially exceeding £16 billion—could, over time, further reduce retrofitting costs for homeowners as the increasing adoption of PV technology drives economies of scale and fosters technological advancements [90]. Effective policies in the past have proven to be possible to positively affect industry prices and end-user implementation of green innovations. Indeed, uptake of biogas in countries like India and China have increased as a result of up-to-date policy initiatives [99]. Therefore, an effective and demographically tailored regulation on retrofitting schemes would significantly decrease household renovation prices for households while also enabling focused investments in green technologies. Further decreasing prices and enabling more properties to reach higher energy efficiencies by reducing financial limits to MEES and making EPC C potentially more affordable under an elemental approach.

Only 29% of retrofitting costs would fall below current energy retrofitting household spending of £3,500 as currently implemented by the Private Renting Energy Efficiency Regulations [69]. Hence, as suggested by the 2021 amendment proposal [5], an increase in household spending budget would be required to require the desired number of properties to reach an EPC of C. Policy makers should consider tailoring this threshold by property type and construction age. According to the aggregated results therein, the marginal abatement costs for residential emissions can be derived and is estimated at £3,273 to £5,124 per tonne of CO_2 . This is significant compared to system-wide abatement costs estimated from changing fuel-source of the UK electricity grid [100]. Suggesting placing all retrofitting costs on end-users to be less efficient than a universal carbon tax to decarbonize the grid and highlighting the importance of introducing system-wide policies in parallel to increasing MEES.

Several limitations underpin the findings and recommendations suggested above. First, EPCs have been proven as better estimators of energy cost-efficiency [58] and tend to overestimate energy use [63]. Therefore, more accurate indicators of energy use such as meter data would be preferable to assess post-retrofitting reductions in energy usage. Second, certain types of retrofits are more intrusive than others and require longer installation times resulting in higher costs, further disincentivizing landlords to commit to these in fear of loss of rental income. Third, the green premium was estimated using a hedonic price model which suffers from different confounding factors, such as demand and supply forces surrounding the point of sale which could result in overestimating the green premium associated with green retrofits. This however provides a ceiling for the estimates in this study. Last, green retrofits might be less costly when committed together as installation costs were computed assuming independent deployment. Improvements, such as cavity wall installations, have also shown not to have long-lasting effects on energy efficiency [101]. By not taking into account such time-varying effects the estimated gains from future energy savings might be overestimated.

Future research may want to look further into each aspect of the effects induced by imposed MEES. These may include looking at the specific improvements undertaken by household characteristics and demography to better understand the national-scale investments needed to deploy effective retrofitting strategies. Additionally, while this study could not rely on tenure statistics for more granular estimates, a deeper understanding of the underlying decisions to resale may provide further insights into the market shock produced by imposed standards. Limitations

of time and processing power prevented this study from other more robust methods of isolating the effects of certain energy improvements. A potential avenue for investigating such trends would have been to use propensity-score matching (PSM) across properties of similar attributes [102, 76, 101, 103, 104].

To be honest I feel that computing a cost of abatement reveals troubling figures, i.e. the cost per ton of carbon avoided is super high... one conclusion would be that it is much cheaper to decarbonise the grid instead, even if the price of electricity increases drastically it nowhere in the region of what a household would pay. this in itself would be incentive enough to focus on the deployment of renewables. boiler upgrade scheme is 7.5k for gov so how many tons avoided from this once you add up consumer costs? (I can give an answer from MCS data BTW)

5. Conclusion

At the national level, mandating improvements in energy efficiency could be pivotal in enabling the UK to meet its Net Zero goals by 2035. The analysis reveals that large-scale investments in photovoltaic (PV) panels, as well as advanced heating and hot water technologies, would be essential for compliance. However, by adopting a strategic approach that leverages learning-by-doing and targeted investments, the UK could substantially reduce the financial burden associated with these upgrades. The study found significant variability in the estimated costs of meeting the proposed energy standards, indicating that more refined and well-communicated strategies could improve the effectiveness and uptake of energy performance improvements beyond the current RdSAP outputs on Energy Performance Certificates (EPCs).

The research corroborates existing empirical evidence on the existence of a green premium for enhanced energy efficiency on property values. While improving properties to an EPC rating of C is likely to have net-positive effects on both property values and energy savings, these benefits must be weighed against the substantial costs of retrofitting. These costs vary significantly depending on the age of the property and fuel poverty in a given area as for older and rural dwellings with low energy efficiency, retrofitting costs would exceed government budgets, impeding the financial viability of these investments. Consequently, landlords of privately rented properties may find it more financially prudent to sell rather than risk exposure to unstable energy and rental markets. The study estimates that approximately 500,000 properties could be at risk of sale due to these pressures, potentially triggering an 11% decline in housing prices within the UK market.

The findings of this study suggest threefold benefits of implementing a minimum EPC rating policy as this single policy could lead to sectoral emissions reductions, create green premiums for homeowners, and induce a significant correction in housing prices. While this could potentially help alleviate the current housing crisis, further research is needed to estimate the drivers behind the decisions of committing to green retrofits and the extent to which certain technologies, such as solar PVs and heat pumps, could bring combined benefits for property owners. A minimum EPC C standard for privately rented homes in England and Wales, when combined with grid-wide decarbonization strategies and tailored subsidy programs, could serve as a comprehensive 'triple-win' strategy, effectively harnessing market dynamics to achieve environmental, economic, and social benefits.

Appendix A. Additional Results

Appendix B. Appendix

Table B.8: Total Improvements to Reach EPC C ordered by Total Costs, according to EPC Recommendation

ID	Count	Description	COST (£)	ΔEPC	Energy(KWh)	CO2 (CO ₂ eq/yr)	TOTAL COSTS (£)
7	3049761	50 mm internal or external wall insulation	9000	6	-0.886	-52.080	27447849000
19	2527465	Solar water heating	5000	3	0.474	-30.535	12637325000
34	2413441	Solar photovoltaic panels, 2.5 kWp	4500	6	-1.469	-27.922	10860484500
58	1939726	Solid floor insulation	5000	4	-0.636	-34.684	9698630000
44	253359	Wind turbine	19999	6	-0.870	-37.588	5067002649
20	1850783	Replace boiler with new condensing boiler	2550	4	-0.752	-33.063	4719589189
8	475419	Replace single glazed windows with low-E double glazing	4900	2	-0.451	-21.554	2329553100
6	2164639	Cavity wall insulation	1000	5	-0.641	-40.683	2164639000
57	1465396	Suspended floor insulation	1000	4	-0.583	-32.612	1465396000
47	1380229	Floor insulation	1000	5	-0.630	-33.051	1380229000
46	486904	Room-in-roof insulation	2100	8	-1.919	-49.068	1022498400
27	188241	Change heating to gas condensing boiler	5000	18	-2.203	-173.602	941205000
11	1831495	Upgrade heating controls	400	2	-0.319	-15.115	732598000
45	520148	Flat roof insulation	1175	4	-0.604	-35.291	611173900
5	1656813	Increase loft insulation to 270 mm	225	3	-0.452	-26.047	372782925
59	230070	High heat retention storage heaters and dual immersion cylinder	1501	17	-0.724	-85.813	345404091
60	210162	High heat retention storage heaters	1346	9	-0.489	-64.509	282899068
48	117052	High performance external doors	1366	5	-0.215	-50.209	159846211
56	101427	Replacement glazing units	1200	1	-0.146	-16.667	121712400
4	404280	Hot water cylinder thermostat	300	4	-0.663	-32.833	121284000
50	149896	Flue gas heat recovery device in conjunction with boiler	650	4	0.562	-42.050	97432400
35	3183726	Low energy lighting for all fixed outlets	30	1	-0.108	-5.385	96785270
30	73306	Fan assisted storage heaters and dual immersion cylinder	1124	16	-0.754	-67.301	82381283
23	7285	Wood pellet stove with boiler and radiators	10000	23	-14.619	-235.002	72850000
25	60639	Fan assisted storage heaters	999	9	-0.751	-58.682	60563201
26	24853	Replacement warm air unit	1875	2	-0.380	-14.726	46599375
63	103238	Party wall insulation	450	3	-0.142	-32.712	46457100
37	17745	Install condensing boiler	2600	14	-3.988	-47.163	46137000
10	409837	Draughtproof single-glazed windows	100	4	-1.012	-28.884	40983700
49	57552	Heat recovery system for mixer showers	655	1	-0.089	-7.041	37696560
22	2609	Replace boiler with biomass boiler	10000	15	-11.784	-106.260	26090000
1	847728	Insulate hot water cylinder with 80 mm jacket	22	4	-0.366	-40.557	19073880
43	863	Condensing oil boiler with radiators	12500	10	-2.715	-18.210	10787500
9	8530	Secondary glazing to single glazed windows	1250	2	-0.661	-12.716	10662500
36	1223	Replace heating unit with condensing unit	2600	3	-0.396	-22.771	3179800

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Table B.9: Total Improvements to Reach EPC C ordered by Total Costs, according to 'Highest Impact on Energy Efficiency' decision model

ID	Count	Description	COST (£)	ΔEPC	Energy(KWh)	CO2 (CO ₂ eq/yr)	TOTAL COSTS (£)
34	7297532	Solar photovoltaic panels, 2.5 kWp	4500	6	-1.469	-27.922	32838894000
7	2808763	50 mm internal or external wall insulation	9000	6	-0.886	-52.080	25278867000
44	482490	Wind turbine	19999	6	-0.870	-37.588	9649462257
58	1089477	Solid floor insulation	5000	4	-0.636	-34.684	5447385000
19	937059	Solar water heating	5000	3	0.474	-30.535	4685295000
20	1466874	Replace boiler with new condensing boiler	2550	4	-0.752	-33.063	3740602044
6	1562244	Cavity wall insulation	1000	5	-0.641	-40.683	1562244000
8	285183	Replace single glazed windows with low-E double glazing	4900	2	-0.451	-21.554	1397396700
46	531882	Room-in-roof insulation	2100	8	-1.919	-49.068	1116952200
27	213075	Change heating to gas condensing boiler	5000	18	-2.203	-173.602	1065375000
47	973416	Floor insulation	1000	5	-0.630	-33.051	973416000
57	631081	Suspended floor insulation	1000	4	-0.583	-32.612	631081000
59	265470	High heat retention storage heaters and dual immersion cylinder	1501	17	-0.724	-85.813	398550111
60	270141	High heat retention storage heaters	1346	9	-0.489	-64.509	363636800
45	227797	Flat roof insulation	1175	4	-0.604	-35.291	267661475
11	500339	Upgrade heating controls	400	2	-0.319	-15.115	200135600
48	140109	High performance external doors	1366	5	-0.215	-50.209	191332850
4	419820	Hot water cylinder thermostat	300	4	-0.663	-32.833	125946000
5	458914	Increase loft insulation to 270 mm	225	3	-0.452	-26.047	103255650
30	89147	Fan assisted storage heaters and dual immersion cylinder	1124	16	-0.754	-67.301	100183399
25	82808	Fan assisted storage heaters	999	9	-0.751	-58.682	82704490
23	7876	Wood pellet stove with boiler and radiators	10000	23	-14.619	-235.002	78760000
50	96199	Flue gas heat recovery device in conjunction with boiler	650	4	0.562	-42.050	62529350
37	20603	Install condensing boiler	2600	14	-3.988	-47.163	53567800
56	36168	Replacement glazing units	1200	1	-0.146	-16.667	43401600
10	352144	Draughtproof single-glazed windows	100	4	-1.012	-28.884	35214400
22	3147	Replace boiler with biomass boiler	10000	15	-11.784	-106.260	31470000
49	46446	Heat recovery system for mixer showers	655	1	-0.089	-7.041	30422130
35	751259	Low energy lighting for all fixed outlets	30	1	-0.108	-5.385	22838274
26	9942	Replacement warm air unit	1875	2	-0.380	-14.726	18641250
43	1064	Condensing oil boiler with radiators	12500	10	-2.715	-18.210	13300000
1	380070	Insulate hot water cylinder with 80 mm jacket	22	4	-0.366	-40.557	8551575
9	6800	Secondary glazing to single glazed windows	1250	2	-0.661	-12.716	8500000
63	11592	Party wall insulation	450	3	-0.142	-32.712	5216400
36	411	Replace heating unit with condensing unit	2600	3	-0.396	-22.771	1068600

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Table B.10: Total Improvements to Reach EPC C ordered by Total Costs, according to 'Lowest Price Retrofit' decision model

ID	Count	Description	COST (£)	ΔEPC	Energy(KWh)	CO2 (CO ₂ eq/yr)	TOTAL COSTS (£)
34	3949220	Solar photovoltaic panels, 2.5 kWp	4500	6	-1.469	-27.922	17771490000
7	1152563	50 mm internal or external wall insulation	9000	6	-0.886	-52.080	10373067000
19	1186128	Solar water heating	5000	3	0.474	-30.535	5930640000
20	2236165	Replace boiler with new condensing boiler	2550	4	-0.752	-33.063	5702332558
58	816618	Solid floor insulation	5000	4	-0.636	-34.684	4083090000
44	156729	Wind turbine	19999	6	-0.870	-37.588	3134470290
8	510330	Replace single glazed windows with low-E double glazing	4900	2	-0.451	-21.554	2500617000
6	2187275	Cavity wall insulation	1000	5	-0.641	-40.683	2187275000
57	1876943	Suspended floor insulation	1000	4	-0.583	-32.612	1876943000
47	1578817	Floor insulation	1000	5	-0.630	-33.051	1578817000
11	2884072	Upgrade heating controls	400	2	-0.319	-15.115	1153628800
46	493401	Room-in-roof insulation	2100	8	-1.919	-49.068	1036142100
27	183775	Change heating to gas condensing boiler	5000	18	-2.203	-173.602	918875000
45	489114	Flat roof insulation	1175	4	-0.604	-35.291	574708950
5	1871533	Increase loft insulation to 270 mm	225	3	-0.452	-26.047	421094925
59	245618	High heat retention storage heaters and dual immersion cylinder	1501	17	-0.724	-85.813	368746303
60	238595	High heat retention storage heaters	1346	9	-0.489	-64.509	321172730
56	242748	Replacement glazing units	1200	1	-0.146	-16.667	291297600
48	177252	High performance external doors	1366	5	-0.215	-50.209	242055331
50	271230	Flue gas heat recovery device in conjunction with boiler	650	4	0.562	-42.050	176299500
35	5700005	Low energy lighting for all fixed outlets	30	1	-0.108	-5.385	173280152
4	550664	Hot water cylinder thermostat	300	4	-0.663	-32.833	165199200
30	76695	Fan assisted storage heaters and dual immersion cylinder	1124	16	-0.754	-67.301	86189841
25	75573	Fan assisted storage heaters	999	9	-0.751	-58.682	75478534
49	99023	Heat recovery system for mixer showers	655	1	-0.089	-7.041	64860065
10	620097	Draughtproof single-glazed windows	100	4	-1.012	-28.884	62009700
23	5942	Wood pellet stove with boiler and radiators	10000	23	-14.619	-235.002	59420000
26	29594	Replacement warm air unit	1875	2	-0.380	-14.726	55488750
63	122865	Party wall insulation	450	3	-0.142	-32.712	55289250
37	19364	Install condensing boiler	2600	14	-3.988	-47.163	50346400
1	1148041	Insulate hot water cylinder with 80 mm jacket	22	4	-0.366	-40.557	25830922
22	1798	Replace boiler with biomass boiler	10000	15	-11.784	-106.260	17980000
9	10511	Secondary glazing to single glazed windows	1250	2	-0.661	-12.716	13138750
43	598	Condensing oil boiler with radiators	12500	10	-2.715	-18.210	7475000
36	1324	Replace heating unit with condensing unit	2600	3	-0.396	-22.771	3442400

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Figure 2: Average Yearly Costs by Improvement ID, extracted from EPC recommendations for all dwellings let, sold or retrofitted between 2008 - 2023

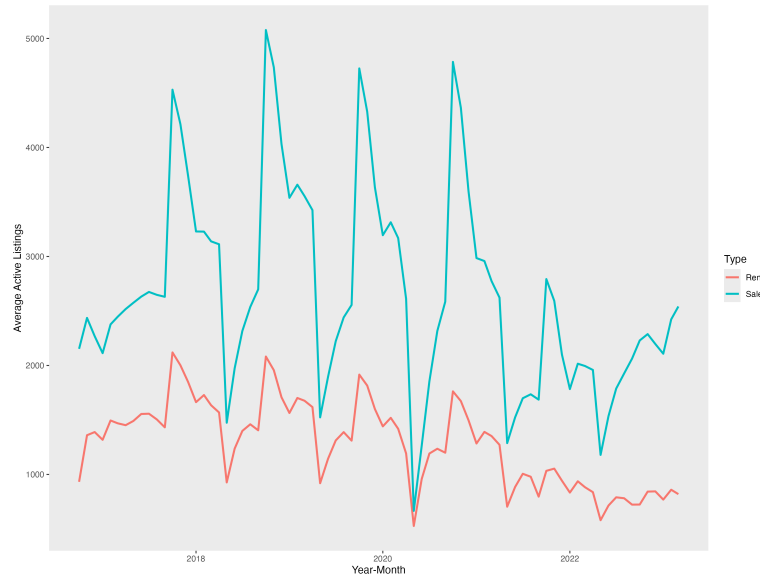


Figure 3: Average monthly rental and sale active listings in UK and Wales.

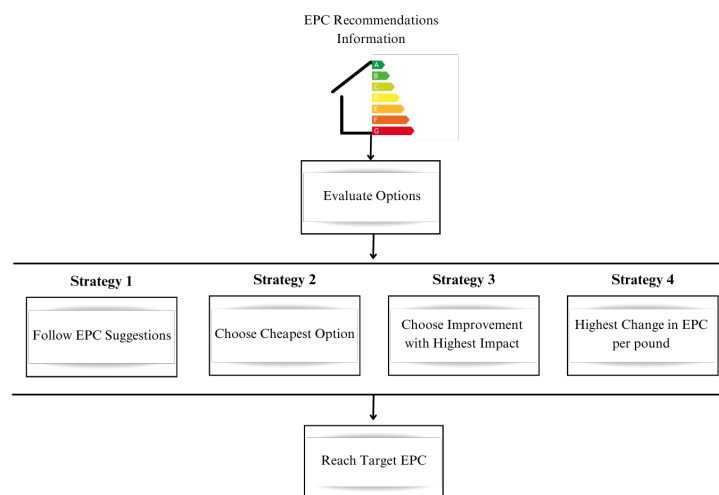


Figure 4: Conceptual framework of decision models used to estimate cost of reaching minimum energy efficiency. The strategies were executed as separate rule-based algorithms applied to the latest energy performance certificates for every unique properties.

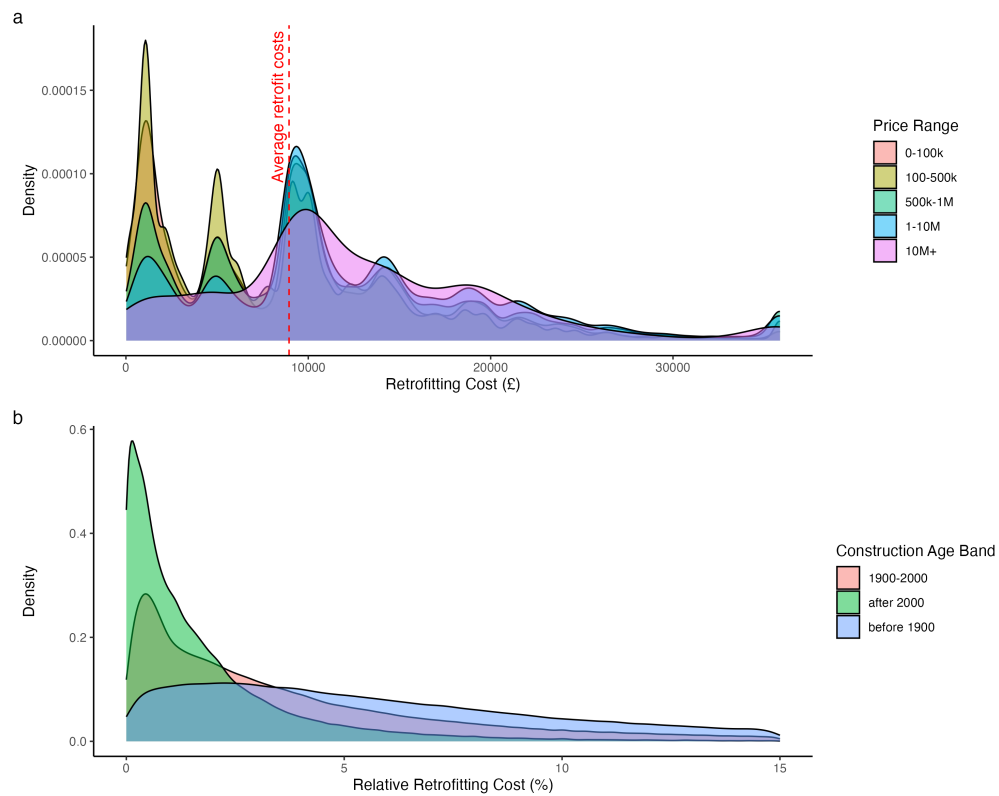


Figure 5: Density plot of absolute (a.) and relative (b.) household retrofitting costs from following EPC recommendations (Strategy 1) to reach hypothetical MEES of EPC - C. Relative costs were determined as percent of property value as of latest transaction. The dashed line in subplot a. represents the average retrofitting cost per household according to this strategy (£9,058 according to Table 3). Top 1% of properties with high retrofitting costs were excluded (winsorized at 99%) to avoid outliers.

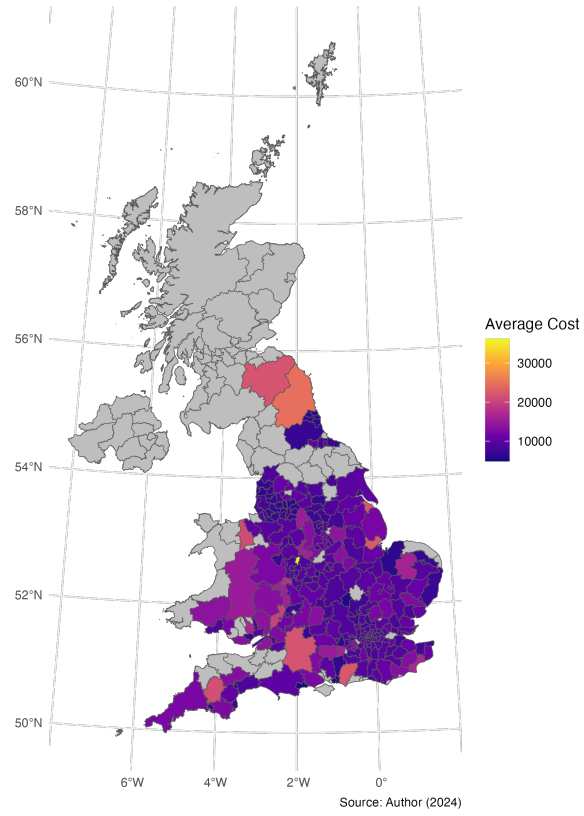


Figure 6: Average retrofitting costs per district in England and Wales (sourced shapesfiles from ONS)

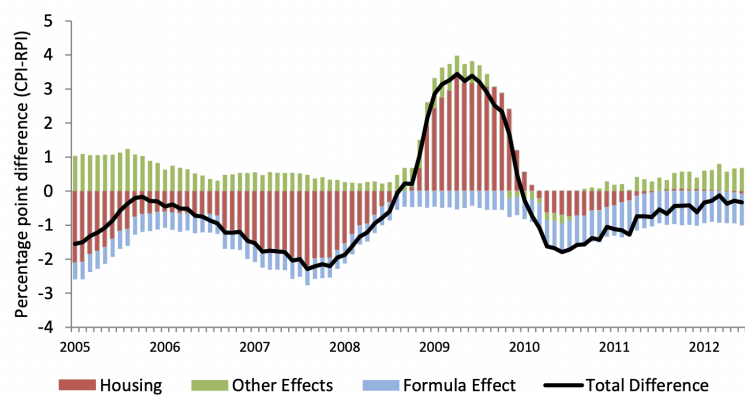


Figure B.7: Contributions of differences between RPI and CPI between 2005-2012. (Source: [72])